

FretGarden

Pedagogical Constitution

Version 1.0 - Foundation

Status	Foundational Draft
Document Type	Pedagogical Standard
Authority	Governing educational specification for FretGarden
Scope	Foundational principles, definitions, constraints, and constitutional success criteria

The educational operating system for every FretGarden lesson, exercise, assessment, explanation, interaction, and future feature.

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1. Preface

FretGarden requires a stable educational identity.

Without an explicit pedagogical constitution, educational products tend to become internally inconsistent as they grow. Individual lessons may be accurate while collectively expressing incompatible assumptions about learning. One lesson may reward speed, another accuracy. One may treat theory as terminology to memorize, another as a way to explain sound. One feature may encourage deliberate practice, while another rewards rapid completion. Over time, the product ceases to feel like the work of one coherent instructor and begins to feel like a collection of disconnected exercises.

This constitution exists to prevent that outcome.

It establishes the principles by which educational decisions in FretGarden are to be made. It governs not only written lessons, but also exercises, assessments, feedback, practice recommendations, sequencing decisions, remediation, review, examples, explanations, progress standards, and any future feature that materially affects learning.

This document is not a curriculum. It does not prescribe the complete order in which every musical topic must be taught. It is not a textbook. It does not attempt to provide complete instruction in music theory, guitar technique, rhythm, composition, improvisation, or ear training. It is not an implementation specification. It does not define interfaces, software architecture, databases, content schemas, or application frameworks except where a technological decision has direct pedagogical consequences.

Instead, this constitution defines the educational rules under which curricula, lessons, systems, and features must be created.

Its purpose is comparable to that of a language specification or engineering standard. It establishes shared definitions, normative requirements, permissible variation, prohibited practices, and the reasoning behind them. Future contributors should be able to consult this document when deciding not merely whether a lesson is factually correct, but whether it belongs in FretGarden.

The constitution is intended to remain durable while allowing the educational system built beneath it to evolve. Specific examples, instructional techniques, assessment formats, and curriculum structures may change as evidence and experience improve. The underlying commitments of FretGarden should change only through deliberate revision of this constitution.

1. Every educational artifact in FretGarden **MUST** conform to this constitution.
2. When a curriculum document conflicts with this constitution, the constitution takes precedence.
3. When an implementation decision conflicts with a pedagogical requirement, the pedagogical requirement takes precedence unless compliance is temporarily impossible.
4. Temporary noncompliance **MUST** be documented as educational debt rather than silently accepted as a new standard.
5. Proposed changes to the constitution **MUST** be evaluated for their effects across the entire educational system, not merely for their local usefulness.
6. Convenience, novelty, engagement metrics, or implementation simplicity **MUST NOT** independently justify violating a constitutional principle.

The standard of quality established here is intentionally higher than factual correctness.

A lesson may contain no factual errors and still be pedagogically unacceptable. It may introduce terminology before the learner has a meaningful experience to attach it to. It may permit advancement without retrieval. It may reward guessing. It may isolate theory from sound. It may teach a scale as a diagram without teaching its interval structure, tonal function, or musical use. It may overwhelm a beginner with information that is accurate but poorly timed.

FretGarden therefore evaluates educational quality across several dimensions:

- factual accuracy;
- conceptual coherence;
- perceptual grounding;
- physical applicability;
- cognitive appropriateness;
- sequencing;
- transfer;
- retention;
- learner agency;
- emotional sustainability;
- and consistency with the larger teaching philosophy.

This constitution treats education as a designed system rather than a collection of explanations. Every local decision creates incentives and expectations. Those incentives accumulate into the learner's understanding of what music is, what practice is, what progress means, and what kind of musician they are capable of becoming.

FretGarden accepts responsibility for those effects.

2. Mission

2.1 Mission Statement

FretGarden's mission is to help learners develop durable, transferable musical understanding through patient, focused, and musically meaningful practice.

FretGarden teaches learners to understand what they hear, recognize what they play, explain why musical patterns work, and apply that knowledge across the fretboard and within real musical contexts.

The mission is not satisfied when a learner can complete an exercise. It is satisfied only when the learner's capability has changed in a way that persists beyond the immediate task.

2.2 Meaning of Durable Understanding

Durable understanding is knowledge or skill that remains available after the original lesson, example, prompt, diagram, or answer choices are removed.

A learner possesses durable understanding when they can retrieve, recognize, explain, perform, and apply a concept under conditions that differ meaningfully from those in which it was introduced.

Durability therefore excludes several forms of apparent success:

- remembering an answer only moments after seeing it;
- reproducing a memorized shape without understanding its musical content;
- identifying a note only when the same diagram is presented;
- performing a passage only at a familiar starting point;
- selecting a correct answer through elimination without being able to produce it independently;
- or completing a unit without retaining prerequisite knowledge.

Future lessons MUST distinguish between temporary task performance and durable learning.

2.3 Meaning of Transferable Understanding

Transferable understanding can be applied beyond a single example. A learner who understands an interval should be able to hear it, locate it, name it, play it from multiple roots, recognize its role in chords and scales, and use it to reason about unfamiliar material. A learner who understands a scale degree should not be limited to one key, one neck pattern, or one exercise format.

Transfer does not require immediate mastery across every context. It does require that instruction be designed toward generalization rather than accidental dependence on one representation. Future lessons MUST therefore identify which parts of an example are essential and which are incidental.

2.4 Mission-Level Educational Commitments

To fulfill its mission, FretGarden MUST:

7. connect theory to sound;
8. connect sound to physical action;
9. connect physical patterns to musical function;
10. require learners to retrieve knowledge rather than merely review it;
11. favor mastery over superficial completion;
12. develop fretboard understanding rather than isolated shape memory;
13. treat rhythm and time as foundational musical skills;
14. build practice habits that are sustainable over long periods;
15. provide meaningful opportunities for application;
16. normalize error as part of learning without normalizing careless repetition;
17. make progress visible without reducing musicianship to points, streaks, or completion percentages;
18. and preserve the learner's ability to make musical choices.

2.5 The Problem the Mission Prevents

Many music-learning systems produce fragmented competence. Learners may know chord diagrams but not chord construction, scale shapes but not scale degrees, note names but not intervals, or theory vocabulary without hearing its musical meaning. They may be able to imitate but not reason, repeat but not transfer, or pass exercises without becoming more independent.

FretGarden's mission is designed to prevent this fragmentation.

How will this help the learner hear, understand, remember, perform, or independently use music?

An educational element that cannot provide a defensible answer SHOULD be removed, revised, or clearly classified as supporting material rather than core instruction.

3. Vision

3.1 Long-Term Vision

FretGarden should become a coherent environment in which learners gradually develop musicianship through repeated contact with sound, structure, movement, reflection, and application.

A learner using FretGarden over an extended period should not experience the product as a sequence of unrelated topics. The learner should experience a progressively expanding network of relationships.

Notes become intervals. Intervals become scale structures and chord structures. Scale degrees acquire tonal tendencies. Chord tones become navigational landmarks. Rhythmic subdivisions become physical and audible. Fretboard locations cease to be isolated coordinates and become manifestations of recurring musical relationships.

The desired result is not merely that the learner knows more facts. The desired result is that music becomes more intelligible.

3.2 The Five-Year Coherence Standard

Five years after this constitution is adopted, a newly written lesson should remain recognizably part of the same educational system as the earliest lessons.

This does not require identical wording or format. It requires continuity of judgment.

A future contributor should reach similar conclusions about when to introduce terminology, how to balance explanation and practice, when a learner is ready to advance, how errors should be treated, what counts as mastery, when visual patterns are useful, when a concept must be heard, how much complexity to introduce, and how a lesson should connect to prior knowledge.

The constitution therefore seeks consistency of reasoning rather than uniformity of presentation.

3.3 Vision of the Learner

FretGarden does not define successful learners by speed, prestige, genre knowledge, technical display, or comparison with others.

A successful FretGarden learner increasingly becomes able to orient themselves on the fretboard; hear and recognize musical relationships; name and explain those relationships; execute them with appropriate rhythm and control; apply them in unfamiliar keys and positions; diagnose their own errors; design useful practice; make intentional musical choices; and continue learning with decreasing dependence on external instruction.

The ultimate direction of the system is learner independence. Instruction that creates permanent dependence on prompts, diagrams, prescribed fingerings, or constant correction conflicts with this vision.

3.4 Vision of Practice

Practice in FretGarden is not defined as repeated activity. It is defined as deliberate activity intended to produce a specific change.

The learner should gradually understand that effective practice requires a clear target, focused attention, an appropriate level of difficulty, accurate repetition, feedback, correction, recovery after error, and sufficient rest.

FretGarden should cultivate the ability to practice, not merely provide material to practice. Lessons MUST teach both musical content and the process by which that content can be learned.

3.5 Vision of Theory

Music theory in FretGarden is not a detached body of labels. Theory is a system for describing, organizing, predicting, and communicating musical relationships.

It should help the learner answer: What am I hearing? Where can I find it? Why does this sound stable or unstable? How is this chord constructed? How does this note function in the key? What changes when the root changes? Why does this pattern recur elsewhere? What options does this structure make available?

Theory MUST explain musical experience. It MUST NOT replace musical experience.

4. Educational Philosophy

4.1 Overview

FretGarden's educational philosophy is built on a hierarchy of meaningful musical experience, attentive perception, accurate physical interaction, conceptual recognition, precise language, deliberate retrieval, varied application, and reflection and integration. This hierarchy is not a rigid lesson template. It establishes dependencies: words should refer to something the learner can increasingly hear, perceive, imagine, locate, or perform.

4.2 Playing Before Abstraction

Whenever reasonably possible, learners SHOULD encounter a musical relationship before receiving an extensive abstract explanation of it. Terminology without experience is cognitively fragile. Concise orientation is permitted, but verbal explanation must not substitute for contact with the musical phenomenon. Lessons should generally move from hearing or playing, through noticing and naming, to explanation, retrieval, and varied use.

4.3 Sound Before Symbol

Musical symbols, diagrams, grids, notation, tablature, interval numbers, scale-degree labels, and chord names represent sound and musical organization. They are not the music itself. A learner MUST be directed back toward sound whenever a representation risks becoming self-contained.

4.4 Theory Always Explains Sound

Every theoretical concept taught as core content **MUST** be connected to an audible consequence or musical use. Theoretical detail that does not yet serve an audible or practical purpose **MAY** be deferred. This prevents inert knowledge: information the learner can repeat but cannot use.

4.5 Relationships Before Isolated Facts

Music is relational. Notes derive meaning from their relationship to other notes, a tonal center, a chord, a rhythmic position, and surrounding context. FretGarden **MUST** avoid teaching isolated facts as though isolation were the final form of understanding.

4.6 Retrieval Over Rereading

Learners strengthen memory by attempting to recall information, not merely by encountering it repeatedly. FretGarden **MUST** include retrieval demands and distinguish recognition, cued recall, free recall, reconstruction, performance, and application.

4.7 Mastery Over Completion

Completion records exposure. It does not necessarily record learning. FretGarden **MUST NOT** treat the end of a lesson or unit as proof of mastery. Where strict mastery is not required, the system **MUST** distinguish introduced, developing, functional, secure, and fluent knowledge.

4.8 Accuracy Before Speed

Speed amplifies what has already been practiced. FretGarden **MUST** treat control, timing, clarity, and repeatability as prerequisites for meaningful tempo increases. Speed increases must be earned by stability rather than used to conceal instability.

4.9 Consistency Over Intensity

Musical development depends on repeated contact over time. A ten-minute focused session may be educationally superior to an hour of distracted repetition. Practice guidance **MUST** avoid implying that meaningful progress requires unusually long sessions.

4.10 Patience Over Premature Advancement

Learning produces delayed and uneven results. FretGarden **MUST** not create pressure to advance solely to preserve momentum or engagement. It should help learners tolerate productive difficulty and return to developing material without framing review as failure.

4.11 Productive Difficulty

Practice must be difficult enough to require attention but not so difficult that success depends primarily on guessing or repeated failure. Difficulty **MUST** serve learning objectives and **MUST NOT** be increased merely to make an activity feel demanding.

4.12 Integration of Ear, Mind, and Hand

FretGarden treats musical learning as coordination of perception, understanding, and action. These domains may be isolated temporarily but **MUST** be reintegrated. Curriculum decisions **SHOULD** identify which domains are being developed and when they will reconnect.

4.13 Reflection as Part of Practice

Learners must gradually learn to observe the quality of their own practice: target, improvement, instability, cause of error, appropriate tempo, next repetition, and effective strategy. Reflection should remain concise enough to support practice rather than displace it.

5. Core Values

5.1 Patience

Patience is the willingness to allow capability to develop through repeated, appropriately difficult practice without demanding immediate fluency.

Why it exists

Musical learning contains unavoidable delay; systems that treat delay as failure encourage rushing and abandonment.

Governing rule

FretGarden **MUST** communicate that time and repetition are normal while still requiring attentive effort and correction. Patience **MUST NOT** excuse vague instruction or ineffective practice.

Effect on future lesson design

Lessons should permit repeated attempts, distinguish slow development from careless repetition, avoid shame, provide actionable correction, revisit important material, and make incremental improvement observable.

5.2 Consistency

Consistency is the repeated return to deliberate practice across time.

Why it exists

Memory and physical skill are strengthened by spaced contact; excessive demands create cycles of overexertion and avoidance.

Governing rule

FretGarden **MUST** favor practice structures that a learner can realistically repeat.

Effect on future lesson design

Lessons should support meaningful work at multiple session lengths, including focused ten-minute sessions.

5.3 Precision

Precision is the accurate use of musical concepts, language, examples, feedback, and success criteria.

Why it exists

Small ambiguities compound and create misconceptions that become difficult to correct.

Governing rule

FretGarden MUST use terminology consistently and distinguish commonly conflated concepts.

Effect on future lesson design

Every lesson should define new terms, preserve established definitions, and avoid casual wording that contradicts technical meaning.

5.4 Musical Meaning

Musical meaning is the audible, structural, expressive, or practical significance of what is being learned.

Why it exists

Learners are unlikely to retain or transfer information that exists only as an arbitrary fact.

Governing rule

Core instruction MUST explain why a concept matters musically.

Effect on future lesson design

Examples should demonstrate function, and exercises should lead toward hearing, playing, analysis, creation, or musical decision-making.

5.5 Agency

Agency is the learner's growing ability to make informed musical and practice decisions.

Why it exists

A learner can become compliant without becoming independent; excessive prescription weakens decision-making.

Governing rule

FretGarden MUST gradually transfer responsibility from the lesson to the learner.

Effect on future lesson design

Later lessons should increasingly require learners to select positions, roots, fingerings, tempos, strategies, or musical applications.

5.6 Honesty

Honesty is the accurate representation of progress, difficulty, mastery, and uncertainty.

Why it exists

False praise and inflated progress claims damage trust and self-assessment.

Governing rule

FretGarden MUST be encouraging without misrepresenting performance.

Effect on future lesson design

Feedback should identify what succeeded, what remains unstable, and what to do next. Praise should refer to observable behavior or improvement.

5.7 Sustainability

Sustainability is the ability to continue learning without unnecessary physical, cognitive, or emotional exhaustion.

Why it exists

A system that produces rapid activity but frequent burnout is educationally unsound.

Governing rule

FretGarden **MUST** treat attention, rest, physical comfort, and workload management as pedagogical concerns.

Effect on future lesson design

Practice recommendations should account for session length, breaks, repetition quality, tension, and diminishing attention.

5.8 Respect

Respect is the treatment of learners as capable people engaged in a demanding craft.

Why it exists

Beginners lack knowledge, not intelligence. Condescension, manipulation, and shame undermine serious learning.

Governing rule

FretGarden **MUST** remain patient, direct, and respectful.

Effect on future lesson design

Explanations should be accessible without becoming simplistic; the garden metaphor must not become sentimental or childish.

5.9 Coherence

Coherence is the alignment of lessons, exercises, assessments, practice guidance, terminology, and progression.

Why it exists

Contradictory incentives teach contradictory values.

Governing rule

No educational feature may be evaluated only in isolation. Its effects on the larger learning system **MUST** be considered.

Effect on future lesson design

An assessment must not reward rapid guessing after a lesson teaches slow retrieval; answers must not remain permanently visible in a retrieval-based system.

5.10 Musicianship Over Task Performance

Musicianship is the integrated ability to hear, understand, perform, create, and make musical decisions.

Why it exists

Educational products can optimize learners for internal task formats rather than for music.

Governing rule

FretGarden MUST evaluate whether skills transfer beyond its immediate prompts and representations.

Effect on future lesson design

Assessments and exercises should vary context and require performance, reconstruction, prediction, or application beyond selecting a displayed answer.

6. Definitions and Terminology

6.1 Normative Language

MUST indicates a binding requirement. MUST NOT indicates a prohibited practice. SHOULD indicates a strong expectation from which departure requires a specific pedagogical justification. SHOULD NOT identifies a generally incompatible practice. MAY identifies a permitted option. Core content is knowledge or skill expected to be retained, retrieved, and used later. Supporting content assists orientation, context, or enrichment but is not necessarily required for progression.

6.2 Learner and Student

Learner and student refer to the person receiving instruction. Learner is preferred when discussing agency and active development. Neither term implies age, prior education, genre preference, or professional aspiration.

6.3 Lesson

A lesson is a structured educational unit intended to produce one or more defined changes in knowledge, perception, physical ability, judgment, or practice behavior. Every lesson MUST have a learning objective, appropriate prerequisites, guided contact with the target, an opportunity for learner action, feedback or self-evaluation, and a defined relationship to later learning.

6.4 Exercise

An exercise is a bounded activity designed to strengthen or assess a specific component of learning. It is not automatically educational because it requires effort; it must have a clear target and produce interpretable evidence.

6.5 Drill

A drill is a deliberately constrained exercise emphasizing repeated retrieval, recognition, movement, timing, or execution. Drills are permitted and often necessary, but become weak when their relationship to musical understanding is absent or permanently deferred.

6.6 Practice

Practice is intentional activity undertaken to produce a change in capability. Mere repetition is insufficient. Practice requires a target, attention, feedback, and adjustment.

6.7 Performance

Performance is the execution of a skill under specified conditions. Good performance during a single attempt does not necessarily prove learning.

6.8 Learning

Learning is a relatively durable change in capability. Evidence requires retention, transfer, or both.

6.9 Mastery

Mastery is the demonstrated ability to perform or apply a defined skill with sufficient accuracy, stability, understanding, and independence for its intended future use. Every mastery claim **MUST** specify scope and conditions.

6.10 Fluency

Fluency is reliable access or execution with reduced conscious effort while preserving accuracy and adaptability. Fluency is not identical to speed.

6.11 Accuracy

Accuracy is the degree to which a response, explanation, perception, or performance matches the intended target.

6.12 Stability

Stability is the ability to reproduce a result across multiple attempts and minor changes in context. A single correct response is not stable evidence.

6.13 Retention

Retention is the availability of learning after a meaningful delay.

6.14 Transfer

Transfer is the successful use of learning in a context that differs from the original instruction.

6.15 Retrieval

Retrieval is the attempt to access knowledge without first being shown the complete answer.

6.16 Recognition

Recognition is identifying the correct concept, sound, answer, or pattern when encountered. It is generally less demanding than recall and **MUST NOT** be treated as equivalent evidence.

6.17 Recall

Recall is producing information from memory through naming, locating, constructing, explaining, playing, singing, counting, or predicting.

6.18 Perception

Perception is the attentive detection or discrimination of musical features.

6.19 Audiation

Audiation is the internal hearing or anticipation of music when the sound is not externally present.

6.20 Fretboard Fluency

Fretboard fluency is the ability to navigate and understand the fretboard through note identity, interval relationships, scale degrees, chord tones, recurring structures, and musical function. It is not equivalent to memorizing isolated diagrams.

6.21 Shape

A shape is a visible or physical configuration on the fretboard. Shapes are useful representational tools but are not complete musical explanations.

6.22 Pattern

A pattern is a recurring organization that may be visual, physical, auditory, rhythmic, intervallic, harmonic, or conceptual. Pattern **MUST NOT** be used as an imprecise substitute for shape.

6.23 Root

A root is the reference pitch from which a chord, interval, or scale construction is named or organized. It is not always the lowest sounding note.

6.24 Tonic

The tonic is the tonal center of a key or tonal context. Root and tonic **MUST NOT** be treated as interchangeable.

6.25 Scale Degree

A scale degree is a note's ordered position and functional relationship within a scale or tonal context.

6.26 Chord Tone

A chord tone is a pitch belonging structurally to a chord under the analytical framework being used.

6.27 Tempo

Tempo is the rate of the underlying beat, ordinarily measured in beats per minute. Tempo is not rhythm.

6.28 Beat

The beat is the recurring pulse used to organize musical time.

6.29 Rhythm

Rhythm is the organization of durations, attacks, rests, accents, and temporal relationships.

6.30 Subdivision

Subdivision is the division of a beat into smaller, regularly related units and must be understood physically and audibly, not merely mathematically.

6.31 Metronome Practice

Metronome practice is deliberate work using a stable external pulse to develop timing, subdivision, coordination, accuracy, tempo awareness, and internal time.

6.32 Productive Error

A productive error is an incorrect attempt that provides interpretable information and is followed by correction.

6.33 Careless Error

A careless error is an incorrect attempt arising primarily from insufficient attention, inappropriate speed, or repetition without adjustment.

6.34 Remediation

Remediation is targeted instruction or practice intended to repair an unstable prerequisite, misconception, or recurring error. It is not punishment and MUST NOT be framed as regression.

6.35 Review

Review is the planned return to previously introduced material for retrieval, strengthening, integration, or transfer.

6.36 Cognitive Load

Cognitive load is the demand placed on limited working attention. Difficulty caused by avoidable wording, irrelevant detail, inconsistent representations, or too many simultaneous demands **SHOULD** be reduced.

6.37 Scaffolding

Scaffolding is temporary support that allows the learner to perform or understand something not yet possible independently. It **MUST** be reduced when no longer needed.

6.38 Feedback

Feedback is information that helps the learner evaluate the relationship between an attempt and the intended objective. It must be sufficiently specific to guide the next attempt.

6.39 Assessment

Assessment is the deliberate collection and interpretation of evidence about learning. It must align with the capability it claims to measure.

6.40 Educational Debt

Educational debt is a known inconsistency, omission, shortcut, or temporary compromise that weakens compliance. It **MUST** be documented with learner impact, scope, and proposed resolution.

7. Design Principles

7.1 Every Lesson Must Produce an Observable Change

A lesson **MUST** identify what the learner should be able to do differently after instruction. Vague objectives produce vague instruction and invalid assessment. Writers must define the intended capability before selecting explanations or activities.

7.2 Prerequisites Must Be Explicit

Every lesson **MUST** identify conceptual, auditory, rhythmic, physical, or navigational abilities it assumes. Readiness checks, review links, or remediation paths are required when prerequisites are essential.

7.3 One Primary Learning Burden at a Time

A lesson SHOULD introduce one principal conceptual or physical burden at a time. Other demands should be familiar or intentionally controlled. Complex tasks should be decomposed and later recombined.

7.4 Decomposition Must Lead Back to Integration

Components may be practiced separately, but isolated practice MUST eventually reconnect to meaningful musical use.

7.5 Examples Must Be Chosen for Educational Function

Examples MUST be selected because they reveal the target relationship clearly. Stylistic appeal may influence selection but cannot replace instructional purpose.

7.6 Multiple Representations Must Converge

Sound, language, fretboard diagrams, tablature, notation, interval formulas, scale degrees, chord symbols, and movement MUST refer to the same underlying relationships and MUST NOT contradict one another.

7.7 Visual Shapes Must Be Explained Relationally

Shapes MAY reduce complexity and support navigation, but MUST NOT be taught as self-justifying objects. Instruction should identify roots, intervals, notes, functions, boundaries, and connections according to level.

7.8 Feedback Must Change the Next Attempt

Feedback MUST provide information the learner can use. Correctness judgments alone are rarely sufficient; feedback should distinguish error types and recommend a next action.

7.9 Assessment Must Match the Claim

An assessment MUST test the capability it claims to measure. Multiple choice cannot independently establish free recall; naming a diagram cannot establish auditory recognition; playing a memorized pattern cannot establish interval understanding.

7.10 Advancement Must Be Evidence-Based

Progression SHOULD occur when the learner has sufficient prerequisite capability, not merely when content has been viewed or attempted.

7.11 Review Must Be Planned, Not Accidental

Core knowledge MUST reappear after its initial lesson using spacing, retrieval, varied context, and integration.

7.12 Difficulty Must Be Diagnosable

When an exercise becomes harder, the source of added difficulty **SHOULD** be identifiable. Difficulty variables should be altered deliberately.

7.13 Musical Context Must Be Sufficient but Not Distracting

Concepts **SHOULD** appear in contexts that demonstrate use without obscuring the target. Writers should choose the smallest musically credible context, then expand later.

7.14 The Learner Must Eventually Generate

Core instruction **SHOULD** move beyond identification toward locating, constructing, completing, varying, improvising, composing, explaining, predicting, or designing practice.

7.15 Practice Instructions Are Part of the Lesson

A lesson that tells the learner what to learn but not how to practice it is incomplete when continued work is required. Guidance should define target, structure, tempo, attention, error response, and stopping conditions when relevant.

7.16 Explanations Must Be Layered

Initial explanations **SHOULD** provide the minimum complete model needed for the current task. Simplification may omit detail but **MUST NOT** establish a misconception that later lessons must undo.

7.17 Terminology Must Follow Need

A term **SHOULD** be introduced when it gives useful control over a concept the learner has encountered or is about to use.

7.18 Practice Must Include Recovery

Learners **MUST** be taught what to do after an error: stop, identify the cause, reduce tempo, isolate movement, return to an earlier beat, count aloud, simplify, or repeat a correct version.

7.19 Tempo Is a Variable, Not a Score

Tempo **SHOULD** control difficulty and measure stable development. It **MUST NOT** become the sole or universal representation of progress.

7.20 Engagement Must Serve Learning

Engagement techniques **MAY** support attention, persistence, curiosity, or meaningful practice. They **MUST NOT** distort priorities, reward shallow completion, encourage compulsive use, or substitute activity for learning.

8. Educational Non-Negotiables

8.1 Sound Is Required

Core pitch, interval, scale, chord, harmony, melody, and rhythm concepts **MUST** include an audible dimension at an appropriate point.

8.2 Active Learner Response Is Required

Core lessons **MUST** require an action that produces evidence. Passive reading or listening alone is insufficient.

8.3 Retrieval Is Required

Core knowledge **MUST** be retrieved after introduction. Continuous visibility of answers does not satisfy this requirement.

8.4 Delayed Review Is Required

Core knowledge **MUST** be revisited after a meaningful delay.

8.5 Application Is Required

Core concepts **MUST** eventually be used outside the exact example in which they were taught.

8.6 Rhythm Cannot Be Treated as Secondary

Rhythm, pulse, subdivision, timing, and temporal control **MUST** be integrated throughout FretGarden.

8.7 Shape-Only Teaching Is Prohibited

Scales, chords, arpeggios, intervals, and fretboard patterns **MUST NOT** be taught solely as visual or physical shapes.

8.8 Speed-First Practice Is Prohibited

Learners **MUST NOT** be instructed to increase speed while accuracy, rhythmic stability, or physical control remain substantially unstable.

8.9 Guessing Must Not Be Rewarded as Mastery

Assessment systems **MUST** distinguish knowledge from chance success.

8.10 Completion Must Not Be Labeled Mastery

Viewing, attempting, or finishing content **MUST NOT** independently produce a mastery claim.

8.11 New Jargon Must Be Defined

Every technical term introduced as instructional content **MUST** be defined or linked to an established definition.

8.12 Definitions Must Remain Consistent

A term **MUST NOT** change meaning between lessons without explicit explanation.

8.13 Prerequisite Failure Must Be Remediable

When struggle arises from an unstable prerequisite, FretGarden **MUST** provide or identify targeted review.

8.14 Errors Must Produce Guidance

When practical, feedback **MUST** distinguish among likely error categories rather than merely report failure.

8.15 Practice Must Be Sustainable

Practice guidance **MUST** permit meaningful progress through manageable, focused sessions and **MUST NOT** imply that long duration is the primary evidence of commitment.

8.16 The Garden Metaphor Must Remain Subordinate

The metaphor **MUST** support educational philosophy and **MUST NOT** obscure technical content, replace precise terminology, trivialize difficulty, or become distracting.

8.17 Learners Must Be Treated Respectfully

Instruction **MUST NOT** be condescending, humiliating, manipulative, sarcastic, or infantilizing.

8.18 FretGarden Must Not Manufacture False Urgency

Core learning **MUST NOT** be framed as a race against other learners or arbitrary timelines.

8.19 Explanations Must Not Establish Known Misconceptions

Simplification is permitted. Falsehood presented as foundational truth is not.

8.20 Future Features Are Subject to Pedagogical Review

Any feature affecting practice behavior, progression, feedback, incentives, assessment, discovery, or representations of success **MUST** be evaluated under this constitution.

9. The Garden Philosophy

9.1 Purpose of the Metaphor

The garden metaphor provides a durable model for musical development: growth requires repeated care; visible progress may lag behind effort; skills develop at different rates; neglected knowledge weakens; excessive force can be counterproductive; healthy development needs effort and rest; and small actions accumulate. The metaphor is pedagogical, not decorative.

9.2 Constitutional Meaning of Garden

The garden represents the learner's developing musical system: conceptual knowledge, auditory perception, physical technique, rhythmic control, fretboard orientation, memory, practice habits, musical judgment, and creative agency. It is actual capability, not points, badges, or completed lessons.

9.3 Seeds

A seed represents an introduced concept or initial experience. Planting a seed is not mastery. Initial exposure must not be represented as secure knowledge.

9.4 Roots

Roots represent foundational relationships and prerequisites such as pulse, subdivision, note identity, whole steps and half steps, interval structure, tonal center, and accurate coordination. Returning to roots is maintenance and strengthening, not regression.

9.5 Watering

Watering represents regular, focused practice. A short attentive session can be meaningful watering; a long session of distracted, inaccurate repetition may provide little useful nourishment.

9.6 Sunlight

Sunlight represents meaningful musical context. Technical work becomes more intelligible when used within riffs, melodies, progressions, improvisation, analysis, or composition.

9.7 Soil

Soil represents existing knowledge, habits, physical condition, attention, and environment. The same lesson may produce different results because it enters different soil.

9.8 Pruning

Pruning represents correction, simplification, and removal of ineffective habits: reducing tempo, eliminating unnecessary motion, correcting misconception, narrowing a target, reducing tension, or replacing an unproductive strategy.

9.9 Seasons

Seasons represent fluctuations in focus, time, confidence, health, interest, and visible progress. A sustainable system allows practice intensity to change without abandoning the identity of being a learner.

9.10 Growth Rings

Growth rings represent accumulated capability that may not be visible daily. Delayed assessment and reflection can reveal gradual improvement.

9.11 Weeds

Weeds represent habits or misconceptions that compete with healthy development: guessing, practicing too fast, relying permanently on diagrams, ignoring rhythm, repeating errors, avoiding weak areas, using tension, or confusing familiarity with mastery. Learners themselves are never weeds.

9.12 The Garden Is Not Passive

Growth is gradual but not automatic. Progress requires focused practice, accurate repetition, retrieval, listening, correction, review, rest, and application. Patience without practice produces stagnation; practice without patience produces frustration and burnout.

9.13 Limits on Metaphorical Language

Garden terminology SHOULD clarify learning expectations, practice, review, progress, correction, patience, reflection, and encouragement. It SHOULD NOT replace standard musical terminology.

9.14 Future Lesson Design

Every lesson should be classifiable as planting a new idea, strengthening roots, watering through practice, connecting branches, pruning an error, revisiting a skill, or applying growth in musical context. The classification need not always be shown to learners.

10. How Students Learn in FretGarden

10.1 The Learning Cycle

FretGarden adopts the general cycle Encounter, Attend, Act, Name, Explain, Retrieve, Vary, Apply, Review, and Integrate. Not every topic requires equal time in every phase, but complete learning usually requires the functions represented by the cycle.

10.2 Encounter

The learner encounters a musical phenomenon in an accessible form through hearing, playing, observing, counting, comparing, or noticing.

10.3 Attend

The learner is directed toward the feature that matters. Without directed attention, a learner may perform an example without perceiving the relevant relationship.

10.4 Act

The learner produces a response through playing, singing, counting, tapping, locating, selecting, constructing, predicting, or explaining.

10.5 Name

The learner connects experience to precise terminology. Naming supports communication but follows or accompanies meaningful contact.

10.6 Explain

The learner develops a working model of why the relationship behaves as it does.

10.7 Retrieve

The learner attempts to access knowledge without complete support. Retrieval begins early but remains appropriately scaffolded.

10.8 Vary

The context changes through a new root, string set, key, register, rhythm, starting point, reduced prompting, or representation.

10.9 Apply

The learner uses the concept for performance, analysis, improvisation, composition, arrangement, ear training, or practice design.

10.10 Review

The learner returns to material after time has passed. Review strengthens retrieval and reveals forgetting.

10.11 Integrate

The concept connects to a larger network: intervals to chords, chord tones to improvisation, scale degrees to function, note locations to movable structures, and subdivisions to riffs and syncopation.

10.12 Students Learn Through Contrast

Contrast helps learners perceive boundaries, but it **MUST** be controlled; too many simultaneous contrasts create confusion.

10.13 Students Learn Through Spacing

Important material should recur across time. Review schedules should reflect educational importance and observed stability.

10.14 Students Learn Through Interleaving

Once initial understanding exists, related skills should sometimes be mixed. Interleaving **SHOULD** follow sufficient initial clarity.

10.15 Students Learn Through Generation

Producing an answer strengthens understanding and exposes misconceptions. Learners should construct intervals, locate chord tones, predict scale degrees, complete rhythms, and create examples.

10.16 Students Learn Through Error and Correction

Error is expected in developing work. A useful error is connected to the target, interpretable, followed by feedback, and corrected in a subsequent attempt.

10.17 Students Learn Through Accurate Repetition

Repetition strengthens what is repeated. A smaller number of attentive, correct, rhythmically stable repetitions is generally preferable to many uncontrolled attempts.

10.18 Students Learn Through Explanation

Explaining how a note was found, why a chord contains a tone, how a pattern moves, what caused an error, or what adjustment is needed can reveal gaps recognition tasks conceal.

10.19 Students Learn Through Prediction

Prediction directs attention and develops internal models. It should be followed by verification.

10.20 Students Learn Through Focused Sessions

Twenty-five minutes of work followed by five minutes of rest is a useful standard. Ten minutes of work followed by five minutes of rest is also legitimate. The constitutional rule is intentional focused periods and restorative breaks at a sustainable scale, not adherence to one duration.

10.21 Students Learn Through Metronome Practice

The metronome clarifies time, not merely speed. Learners must understand BPM, what receives the click, subdivision, note placement, manageable tempo, recovery after losing time, and when an increase is justified. The objective is internal time, not permanent dependence on external clicks.

10.22 Students Learn at Uneven Rates

Conceptual, auditory, and physical capabilities develop asynchronously. FretGarden should identify the specific domain that is secure or developing.

10.23 Students Learn by Revisiting

A concept encountered again at a deeper level is not redundant. Early recognition may later acquire construction, function, transfer, analysis, or creative use.

10.24 Students Learn by Becoming Better Practitioners

Learners should increasingly set targets, choose tempos, identify errors, isolate problems, reduce difficulty, track stable improvement, decide when to stop, and choose what to revisit.

11. What FretGarden Is

11.1 A Guided Practice Environment

A combination of instruction, deliberate exercises, review, feedback, and musical application. Guidance should decrease as capability and judgment increase.

11.2 A System for Building Fretboard Understanding

The fretboard is taught through notes, intervals, chord tones, scale degrees, scales, patterns, and musical relationships as a connected system rather than unrelated boxes.

11.3 A Music-Theory Learning System

Theory is taught as an explanation of sound and structure, developing the ability to reason rather than merely repeat terminology.

11.4 A Practice Coach

FretGarden teaches focus, repetition, tempo control, metronome use, error correction, review, reflection, and sustainable workload.

11.5 A Long-Term Learning System

FretGarden is designed around accumulation, retention, and integration across repeated sessions.

11.6 A Bridge Between Theory and Playing

Theoretical explanation should eventually affect hearing, navigation, analysis, practice, performance, improvisation, or creation.

11.7 A System That Uses Drills Purposefully

Drills are appropriate for retrieval, note locations, interval construction, scale-degree identification, chord-tone targeting, subdivision, and controlled execution when they serve larger musical capabilities.

11.8 A Respectful Beginner Entry Point

Foundational ideas should be clear without treating beginners as childish.

11.9 A Useful Environment for Developing Musicians

Intermediate learners may repair gaps, connect shapes to theory, strengthen rhythm, and improve practice without being forced through every beginner lesson chronologically.

11.10 A Coherent Instructor Identity

FretGarden should feel calm, patient, exact, encouraging, practical, musically informed, and honest about difficulty.

12. What FretGarden Is Not

12.1 Not a Content Library Without a Learning Model

A collection of lessons, diagrams, videos, facts, and exercises is not automatically an educational system. Sequence, prerequisites, review, alignment, and transfer are required.

12.2 Not a Shape-Memorization System

Guitar knowledge is not defined as memorizing boxes and forms. Shapes must connect to notes, intervals, functions, sound, and movement.

12.3 Not a Vocabulary Course

The ability to define terms is not sufficient musical understanding.

12.4 Not a Speed-Training Product

Speed is subordinate to control, timing, accuracy, tone, and musical intention.

12.5 Not a Completion Engine

FretGarden does not exist to move learners through the largest possible quantity of content.

12.6 Not a Competitive Ranking System

Success is not defined through comparison with other learners.

12.7 Not an Arbitrary Gamification Layer

Points, streaks, rewards, and levels may not reward behavior that conflicts with mastery, rest, honest self-assessment, or sustainable practice.

12.8 Not a Substitute for Listening

No visual or verbal explanation eliminates the need to hear music.

12.9 Not a Substitute for Playing

Instrumental knowledge must eventually interact with the instrument.

12.10 Not a Substitute for Musical Choice

FretGarden must create room for interpretation, selection, variation, improvisation, composition, and self-directed practice.

12.11 Not a Marketing Presentation

Educational content must not exaggerate benefits, constantly celebrate ordinary navigation, or describe the product instead of teaching.

12.12 Not Childish or Overly Whimsical

The garden identity must not turn serious instruction into novelty.

12.13 Not Needlessly Academic

Technical accuracy does not require inaccessible prose.

12.14 Not Genre-Prescriptive

Foundational relationships must transfer across genres; no genre is the exclusive expression of musicianship.

12.15 Not a Single Correct Fingering Authority

One fingering should not be presented as universally correct when multiple valid choices exist.

12.16 Not a Source of Medical or Therapeutic Claims

FretGarden may encourage breaks and reduced tension but must not diagnose injury or present pedagogical advice as medical treatment.

12.17 Not an Infinite Dependency

The system succeeds when learners become more capable of hearing, reasoning, practicing, and creating independently.

13. Success Criteria for the Entire Constitution

13.1 Internal Consistency

The constitution **MUST** not establish contradictory requirements. Tensions must be resolved explicitly: sound before symbol does not prohibit notation; mastery does not require perfection; consistency does not prohibit long sessions; patience does not excuse ineffective practice; agency does not eliminate scaffolding.

13.2 Operational Clarity

Contributors should be able to translate principles into decisions about objectives, assessment, timing, drills, progression, feedback, and incentives.

13.3 Terminological Stability

Key terms must be defined once, used consistently, and expanded carefully. Future chapters **MAY** refine but **MUST NOT** silently alter definitions.

13.4 Implementation Independence

The constitution should remain valid across changes in software framework, interface, device, format, or delivery technology.

13.5 Curriculum Independence

The constitution should support multiple valid curricula while preventing pedagogically incompatible ones.

13.6 Assessment Validity

The completed constitution must define how evidence supports claims about exposure, recognition, recall, accuracy, stability, retention, transfer, fluency, and mastery.

13.7 Support for Review and Remediation

Forgetting, review, and prerequisite repair must be normal components of learning.

13.8 Support for Learner Diversity

The constitution must accommodate differences in prior knowledge, practice time, physical reach, configuration, learning rate, goals, genre, reading ability, auditory development, and technical experience without eliminating standards.

13.9 Protection Against Educational Drift

The constitution must prevent drift toward shallow engagement, excessive gamification, completion progression, shape-only instruction, jargon-first explanation, speed obsession, and inconsistent definitions.

13.10 Support for Authoring

A lesson writer should be able to derive objective, prerequisites, sequence, practice structure, retrieval demand, review need, application requirement, and assessment evidence.

13.11 Support for Quality Assurance

Reviewers should be able to cite explicit standards rather than rely only on subjective reactions.

13.12 Support for Revision

Future revisions should include version numbers, change summaries, reasons, affected provisions, compatibility implications, and migration expectations.

13.13 Preservation of Instructor Identity

Different authors should remain aligned in expectations, sequencing, treatment of error, terminology, theory-sound relationship, practice philosophy, mastery standards, and tone.

13.14 Observable Learner Outcomes

The governed system should produce stronger fretboard orientation, note and interval knowledge, rhythmic stability, theory-sound connection, retention, transfer, practice decisions, error diagnosis, and musical independence.

13.15 Resistance to Superficial Compliance

Compliance depends on educational function, not labels. An audio clip alone does not make a lesson sound-based; a quiz alone does not make it retrieval-based; another key alone does not prove transfer; a timer alone does not teach practice.

13.16 Constitutional Test Questions

19. What capability is this intended to change?
20. What prior capability does it assume?
21. What does the learner hear, perceive, play, retrieve, explain, or create?
22. What evidence would show that learning occurred?
23. Does the task measure that evidence directly?
24. What prevents guessing or temporary familiarity from appearing as mastery?
25. How will the knowledge be reviewed?
26. How will it transfer beyond this example?
27. What should the learner do after an error?
28. Does the activity reinforce sustainable practice?
29. Does it develop dependence or independence?
30. Does it treat musical relationships accurately?
31. Does it remain consistent with established terminology?
32. Does it serve musicianship or merely application-specific task performance?
33. Would an experienced FretGarden instructor defend this decision using the constitution?

If these questions cannot be answered, the educational design is not ready for final approval.

14. Unresolved Questions

14.1 Target Learner Boundaries

Define minimum reading ability, instrument access, setup instruction scope, advanced endpoint, and specialized pathways.

14.2 Instrument Scope

Determine whether bass, alternate tunings, extended-range instruments, and other fretted instruments require instrument-specific provisions.

14.3 Ear-Training Requirements

Define minimum expectations, singing requirements, response methods, progression standards, and accommodations for hearing differences.

14.4 Standard Notation and Tablature

Determine the required role, order, and depth of each representation and how dependence should be managed.

14.5 Mastery Thresholds

Define task-specific accuracy percentages, response times, repetition counts, and retention intervals.

14.6 Creative Work

Define standards for improvisation, composition, arrangement, originality, and open-ended evaluation.

14.7 Genre and Repertoire Use

Define selection, analysis, adaptation, licensing, and balance across genres.

14.8 Physical Technique

Establish pedagogy for fretting, picking, muting, posture, articulation, movement, and ergonomics.

14.9 Accessibility

Establish formal auditory, visual, motor, cognitive, and language accessibility standards.

14.10 Motivation and Engagement Systems

Define permitted motivational systems, streak behavior, rewards, notifications, and progress visualizations.

14.11 Placement and Diagnostic Assessment

Define placement for experienced learners and methods for diagnosing gaps.

14.12 Use of Automated Evaluation

Define reliability thresholds, disclosure of uncertainty, and circumstances requiring human interpretation.

14.13 Practice Data

Define the educational meaning and ethical use of duration, accuracy, tempo, consistency, and self-reporting.

15. Assumptions Made

15.1 Primary Instrument

FretGarden's primary instrument is guitar, and the fretboard is a central learning environment.

15.2 Primary Knowledge Domains

Core domains include notes, intervals, scale degrees, chord tones, scales, chords, rhythm, fretboard relationships, ear training, and broader structure.

15.3 Beginner Accessibility

Learners may begin with little or no formal theory knowledge.

15.4 Long-Term Use

FretGarden is intended to support learning across months or years.

15.5 Instrument Access

Many core activities assume access to a playable guitar, though some work may occur away from the instrument.

15.6 Variable Practice Time

The system must remain useful during short sessions.

15.7 Active Practice

FretGarden requires active participation rather than operating primarily as a reference library.

15.8 Musical Independence

The desired endpoint is increasing independence from prompts and prescribed answers.

15.9 Garden Metaphor

The metaphor is foundational to the learning philosophy but subordinate to precise instruction.

15.10 Focused Practice

Pomodoro-style work and break intervals are useful models, but no single duration is mandatory.

15.11 Metronome Use

Metronome practice is core to timing, subdivision, controlled execution, and internal time.

15.12 Educational Evolution

Specific curricula and methods may change; the core philosophy remains stable unless evidence justifies revision.

16. Topics Intentionally Deferred to Future Chapters

35. Curriculum sequencing
36. Prerequisite mapping
37. Lesson architecture
38. Exercise design
39. Drill design
40. Assessment validity
41. Mastery and progression
42. Retrieval practice
43. Spaced review
44. Interleaving
45. Remediation
46. Feedback standards
47. Error classification
48. Ear-training pedagogy
49. Rhythm pedagogy
50. Metronome pedagogy
51. Fretboard pedagogy
52. Note-learning standards
53. Interval pedagogy
54. Scale-degree pedagogy
55. Chord-tone pedagogy
56. Scale and mode pedagogy
57. Chord and harmony pedagogy
58. Technique instruction
59. Improvisation
60. Composition and creative work
61. Repertoire and genre examples
62. Practice-session design
63. Pomodoro and focus structures
64. Reflection and practice logging
65. Motivation and engagement
66. Progress representation
67. Accessibility
68. Adaptive instruction
69. Placement and diagnostics
70. Tone and instructional voice
71. Terminology governance
72. Content review and quality assurance
73. Educational analytics
74. Automated feedback
75. Versioning and constitutional amendment
76. Exceptions, waivers, and educational debt
77. Contributor responsibilities
78. Research and evidence standards

17. Required Future Cross-References

17.1 Learning Objectives

Reference Sections 2, 7.1, 10, and 13.16.

17.2 Assessment

Reference Sections 4.6-4.7, 6.8-6.17, 7.8-7.10, 8.3-8.10, and 13.6.

17.3 Fretboard Instruction

Reference Sections 4.3-4.5, 4.12, 6.20-6.22, 7.6-7.7, 8.7, and 11.2.

17.4 Practice Design

Reference Sections 4.8-4.13, 5.1-5.2, 5.7, 6.5-6.6, 6.31-6.33, 7.15, 7.18-7.19, 8.14-8.15, and 10.17-10.24.

17.5 Rhythm and Metronome Instruction

Reference Sections 6.27-6.31, 7.18-7.19, 8.6, and 10.20-10.21.

17.6 Motivation and Engagement

Reference Sections 5.1-5.2, 5.6-5.8, 7.20, 8.15-8.18, 9, and 12.5-12.7.

17.7 Tone and Voice

Reference Sections 5.3, 5.6, 5.8, 8.11-8.12, 8.16-8.17, 9.13, 11.10, and 12.11-12.13.

17.8 Review and Remediation

Reference Sections 4.6-4.7, 4.10, 6.33-6.36, 7.2, 7.10-7.11, 8.4, 8.13, 10.13-10.14, 10.23, and 13.7.

17.9 Constitutional Governance

Reference Sections 1, 5.9, 6.1, 6.40, 8.20, 13.1-13.5, 13.9-13.12, and 13.15.

18. Foundation Conformance Statement

An educational artifact conforms to this foundation only when it:

79. serves a defined musical or learning capability;

80. respects prerequisite relationships;
81. requires active learner engagement;
82. connects theory to sound and musical use;
83. avoids shape-only or terminology-only instruction;
84. includes retrieval and future review for core content;
85. distinguishes completion from learning;
86. uses assessments that match their claims;
87. treats rhythm as foundational;
88. supports accurate and sustainable practice;
89. provides actionable treatment of error;
90. develops transfer and learner independence;
91. uses terminology consistently;
92. applies the garden philosophy without allowing metaphor to replace instruction;
93. and remains consistent with the mission, vision, values, and non-negotiables defined in this document.

Conformance is determined by educational function rather than superficial format.

This foundation is intended to govern every subsequent chapter of the FretGarden Pedagogical Constitution. Later provisions may become more specific. They may not silently weaken these requirements.

END OF FRETGARDEN PEDAGOGICAL CONSTITUTION V1.0 - FOUNDATION